

METHODS AND RESULTS REPORT

First Recorded Tertiary Education of Wits Staff

Record-level classification of domestic and international first-education locations

Interpretation boundary

This analysis describes the country associated with the earliest classified tertiary credential in a public staff-directory record. It does not measure nationality, citizenship, place of birth, ethnicity, or legal immigration status.

University of the Witwatersrand public staff-directory snapshot

Dataset snapshot: September 2026 | Report prepared: 2 September 2026

Abstract

This report describes a record-level analysis of 4,333 profiles in a supplied September 2026 snapshot of the University of the Witwatersrand public staff directory. A deterministic parser selected the earliest recognized tertiary credential from each qualification string, assigned an institution by parenthetical parsing and backward propagation, and mapped the institution to South Africa, one of 43 foreign country categories, or unspecified. Of 3,526 profiles with a resolved country, 3,087 (87.55%) were classified as South African first education and 439 (12.45%) as international. International classifications were concentrated in the United Kingdom and United States and were more prevalent in several senior or research designations than in lecturer designations. Results are descriptive outputs of the supplied rule-based classifier and should not be interpreted as nationality or citizenship estimates.

Key data-quality finding

The supplied draft reported no pre-tertiary schooling mentions. Direct reconciliation found 43 qualification strings (0.99%) containing a schooling term, almost always 'Matric'. The biography field was blank in all 4,333 records. Pre-tertiary schooling is therefore rare and unusable as a systematic origin measure, but not literally absent.

Report scope

Unit of analysis: One public staff-directory profile URL; results are profile-level rather than verified person-level counts.

Primary outcome: Country category associated with the earliest classified tertiary credential.

Analysis type: Descriptive counts and percentages; no inferential or causal analysis.

Files analyzed: wits_staff_directory.csv and wits_staff_education_enriched.csv, together with the supplied Python parsing and scraping scripts.

1. Methods

1.1 Study design and data source

The study used a cross-sectional census of the supplied Wits public staff-directory snapshot. The raw and enriched CSV files each contain 4,333 rows. The enriched file contains 4,333 unique profile URLs and 4,330 unique displayed names across 313 distinct department strings and 204 distinct designation strings. Because displayed names can repeat and no stable personnel identifier was available, the profile URL was treated as the record key.

The live directory was not re-scraped for this report. All results were recomputed from the supplied local snapshot, so they describe that snapshot and may differ from the current public directory.

1.2 Directory extraction and enrichment

The supplied scraper requested paginated directory pages, parsed staff name, designation, department, email, phone and profile URL, tagged records by page and row position, and sorted those coordinates before export. The directory scraper configured a 16-worker thread pool and an HTTP retry strategy for status codes 429, 500, 502, 503 and 504.

A second script requested individual profile pages using a 25-worker pool and attempted to extract qualification text, ORCID links and biography snippets. In the supplied enriched CSV, qualification text was present for 4,089 profiles (94.37%), while ORCID and biography fields were blank for all profiles. Consequently, the present analysis uses directory metadata and qualification strings only.

1.3 First-education classification algorithm

The deterministic parser implemented the following sequence. The description reflects the supplied code rather than an independently validated natural-language processing model.

Table 1. Implemented classification workflow

Step	Operation	Implemented rule
1	Parse parenthetical blocks	A regular expression separated text preceding parentheses from the parenthetical content.
2	Split compound degree strings	Comma-separated degree tokens within a block became individual candidate credentials.
3	Propagate institutions backward	During a reverse scan, a later non-empty parenthetical institution was assigned to preceding candidates with empty institution fields.
4	Select the first recognized credential	The first token matching a configured undergraduate list was selected; otherwise the first parsed token was used.
5	Resolve country	Institution text was matched against South African keywords and foreign institution/country keywords. If the selected institution did not match, the full qualification string was searched as a fallback.
6	Assign classification	Resolved South African records were labeled domestic; other resolved country categories were labeled international; unmatched records were labeled unspecified.

1.4 Outcome definitions

Table 2. Analytic definitions

Measure	Definition
Domestic	First_Education_Country = South Africa.
International	A non-South-African country category was resolved.
Specified / valid	Domestic or international; excludes Unspecified.
Unspecified	No country category was resolved by the supplied rules.
International share	International / (Domestic + International) x 100.

1.5 Descriptive analysis

Counts and percentages were recomputed directly from the enriched CSV. Country distributions use First_Education_Country. Rank analyses use exact designation strings without consolidating spelling or punctuation variants. Department analyses include departments with at least 30 specified records. Continental summaries were derived from an explicit mapping of all 43 foreign country categories in the dataset.

1.6 Pre-tertiary feasibility check

Qualification strings were searched case-insensitively for matric, high school, secondary school, grammar school and primary school terms. The search identified 43 profiles (0.99%). Most entries merely contained the word 'Matric' without naming a school; one explicitly included 'GERMAN SCHOOL'. Because biography snippets were empty in all records and named schools were essentially absent, the snapshot cannot support systematic primary- or secondary-school provenance analysis.

1.7 Quality checks and interpretation safeguards

Table 3. Reconciliation checks

Check	Result
Row reconciliation	Raw and enriched files each contained 4,333 rows.
Record key	All 4,333 profile URLs were unique.
Qualification completeness	4,089 profiles (94.37%) had non-empty qualification text.
Institution completeness	3,670 profiles (84.70%) had a non-empty selected institution string.
Country classification	3,526 profiles (81.38%) received a domestic or international country classification.
Arithmetic	Country, continent, rank and department totals were independently recomputed from the final CSV.

2. Results

2.1 Cohort completeness and overall classification

Of 4,333 profiles, 3,526 (81.38%) received a resolved first-education country. The remaining 807 (18.62%) were unspecified. Among all profiles, 3,087 (71.24%) were classified as South African and 439 (10.13%) as international. Restricting the denominator to specified profiles, the distribution was 87.55% domestic and 12.45% international.

Table 4. Overall first-education classification

Classification	Profiles	% of all profiles	% of specified profiles
South Africa	3,087	71.24%	87.55%
International	439	10.13%	12.45%
Unspecified	807	18.62%	-
Total	4,333	100.00%	100.00%

Classification of 4,333 staff-directory profiles

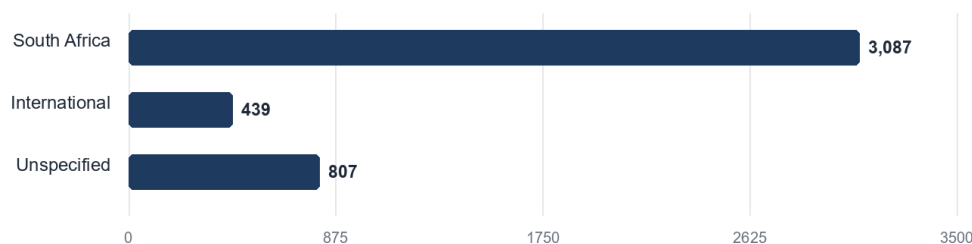


Figure 1. Profile classification in the full directory snapshot.

2.2 International country distribution

The 439 international profiles were distributed across 43 country categories. The United Kingdom accounted for 166 profiles (37.81% of the international group), followed by the United States with 85 (19.36%). Together these two categories represented 57.18% of international classifications.

Table 5. Top 15 international first-education country categories

Country category	Profiles	% international	% specified
UK	166	37.81%	4.71%
USA	85	19.36%	2.41%
Nigeria	18	4.10%	0.51%
DR Congo	12	2.73%	0.34%
Canada	12	2.73%	0.34%
Germany	11	2.51%	0.31%

Country category	Profiles	% international	% specified
France	11	2.51%	0.31%
Australia	10	2.28%	0.28%
Zimbabwe	10	2.28%	0.28%
New Zealand	9	2.05%	0.26%
Netherlands	9	2.05%	0.26%
Belgium	7	1.59%	0.20%
Italy	7	1.59%	0.20%
Mauritius	5	1.14%	0.14%
China	5	1.14%	0.14%

Top 10 international first-education country categories

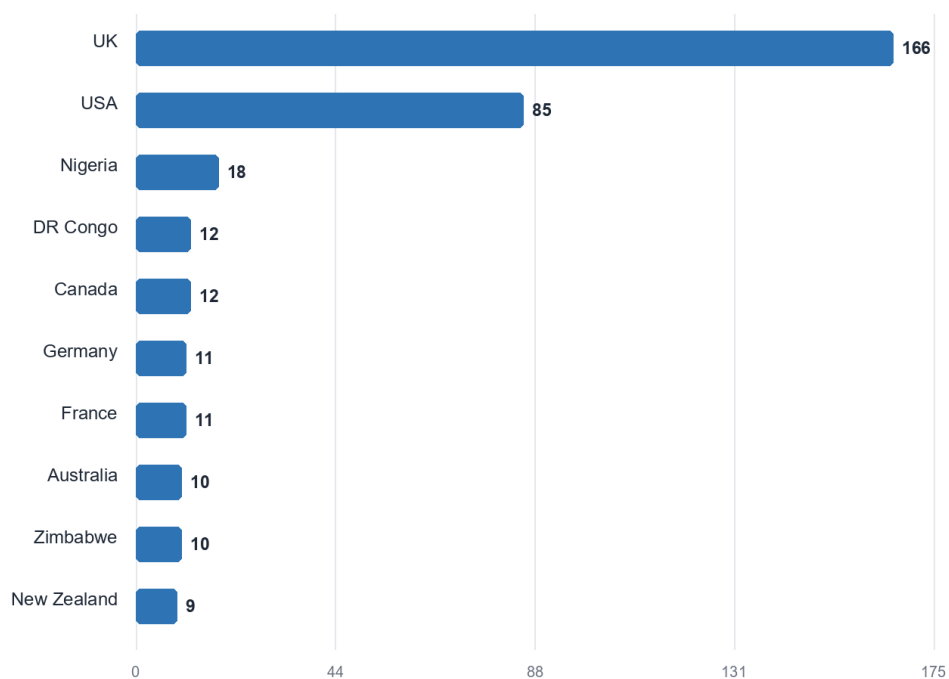


Figure 2. Ten most frequent international country categories.

2.3 Continental distribution

Europe constituted just over half of international classifications. North America was second, while African countries other than South Africa accounted for approximately one in seven international classifications.

Table 6. Continental aggregation of international country categories

Region	Profiles	% international
Europe	235	53.53%
North America	97	22.10%
Africa (excluding South Africa)	65	14.81%
Oceania	19	4.33%
Asia and Middle East	17	3.87%
Latin America	6	1.37%

2.4 Variation by academic designation

International shares differed markedly across selected exact designation labels. Associate Lecturer and Lecturer records had the lowest international shares, whereas Research Associate and Professor records had the highest. These figures are descriptive and do not establish that seniority causes international recruitment; designation labels may also encode different appointment types and have not been normalized.

Table 7. Classification by selected exact designation

Designation	Total	Specified	Domestic	International	% international
Associate Lecturer	944	753	722	31	4.12%
Lecturer	1,254	1,041	992	49	4.71%
Senior Lecturer	415	373	324	49	13.14%
Associate Professor	265	236	195	41	17.37%
Professor	197	176	113	63	35.80%
Researcher	162	131	114	17	12.98%
Research Associate	109	82	44	38	46.34%
Adjunct Professor	57	52	46	6	11.54%

International share within valid records, selected designations

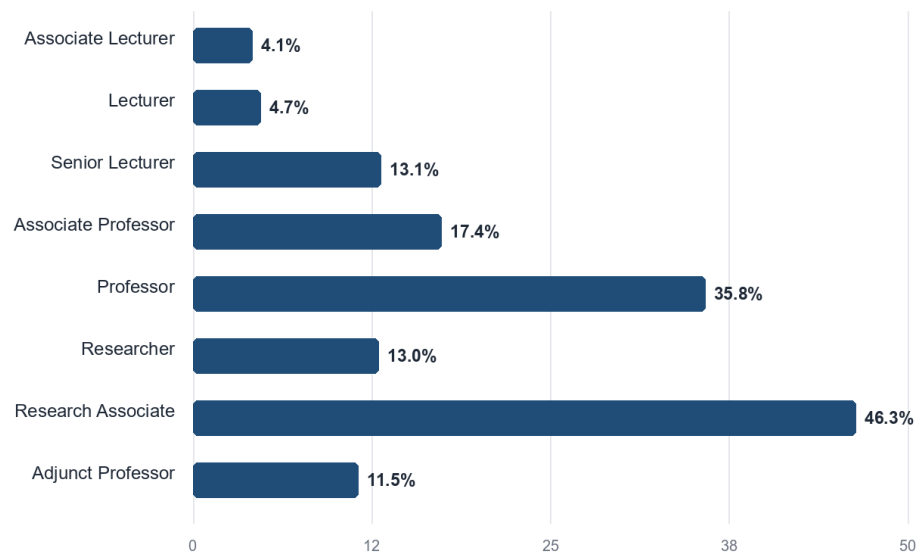


Figure 3. International share within specified records for selected exact designations.

2.5 Variation by department

Among departments with at least 30 specified records, the highest international shares were observed in the Southern Centre for Inequality Studies, Wits School of Governance, School of Physics and School of Geosciences. Several hospital-based clinical departments had international shares below 6%. The data do not directly test the institutional or regulatory mechanisms that produced these patterns.

Table 8. Departments with the highest international shares (at least 30 specified records)

Department	Specified	Domestic	Intl.	Intl. %
Southern Centre for Inequality Studies	41	25	16	39.02%
Wits School of Governance	37	23	14	37.84%
School of Physics	59	38	21	35.59%
School of Geosciences	40	27	13	32.50%
School of Geography, Archaeology and Environmental Studies	33	23	10	30.30%
School of Law	71	52	19	26.76%
School of Economics and Finance	56	42	14	25.00%
School of Animal, Plant and Environmental Sciences	70	55	15	21.43%
School of Public Health	43	34	9	20.93%
School of Computer Science and Applied Mathematics	47	38	9	19.15%
School of Electrical & Information Engineering	32	26	6	18.75%
Graduate School of Business Administration	50	42	8	16.00%

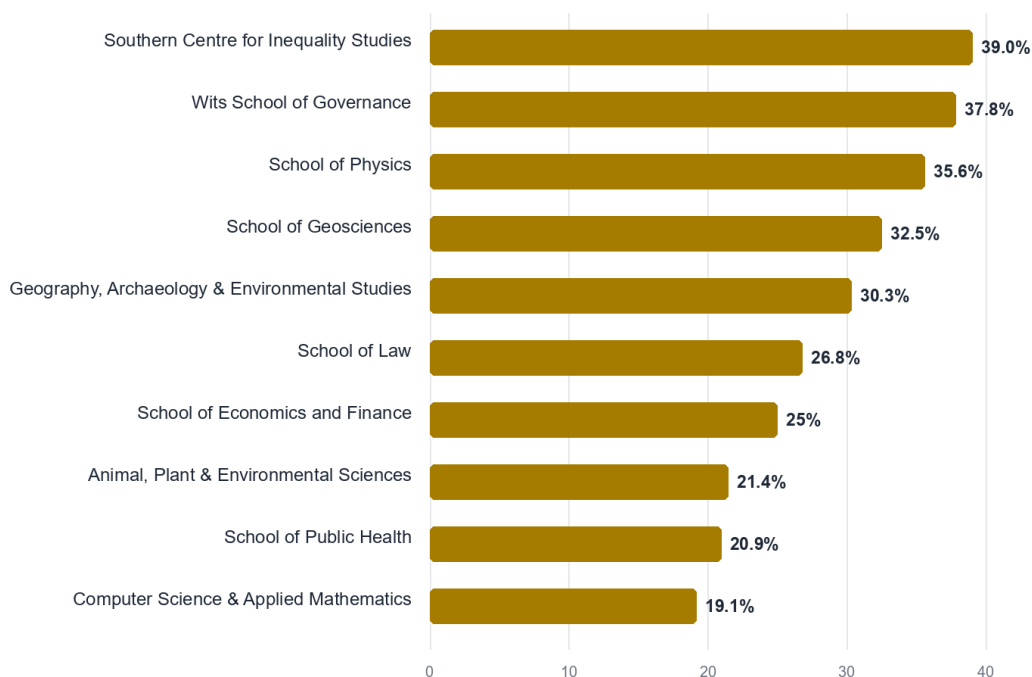
Highest international shares (at least 30 valid records)

Figure 4. Ten departments with the highest international shares among those with at least 30 specified records.

Table 9. Selected clinical departments with low international shares

Department	Specified	Domestic	Intl.	Intl. %
Paediatrics & Child Health (Johannesburg Hospital)	40	40	0	0.00%
Psychiatry (Chris Hani Baragwanath Hospital)	32	32	0	0.00%
Paediatrics & Child Health (Chris Hani Baragwanath Hospital)	63	62	1	1.59%
Internal Medicine (Chris Hani Baragwanath Hospital)	56	55	1	1.79%
Emergency Medicine	43	42	1	2.33%
Anaesthesiology (Johannesburg Hospital)	42	41	1	2.38%
General Surgery (Johannesburg Hospital)	41	40	1	2.44%
Clinical Microbiology And Infectious Diseases	34	33	1	2.94%
General Surgery (Chris Hani Baragwanath Hospital)	32	31	1	3.12%
Anaesthesiology (Chris Hani Baragwanath Hospital)	54	51	3	5.56%
Orthopaedic Surgery (Chris Hani Baragwanath Hospital)	35	33	2	5.71%
Obstetrics & Gynaecology (Chris Hani Baragwanath Hospital)	35	33	2	5.71%
Internal Medicine(Johannesburg Hospital)	55	51	4	7.27%

2.6 Pre-tertiary schooling information

The pre-tertiary keyword search returned 43 profiles (0.99%). The overwhelming majority recorded only 'Matric' and did not identify a school. With no populated biography text and virtually no named schools, the dataset does not permit a reliable school-origin classification. First recorded tertiary education remains the earliest broadly available education field in this snapshot, but it should be described as a pragmatic proxy rather than an objective measure of geographic origin.

3. Limitations

Proxy construct

First recorded tertiary education is not equivalent to nationality, citizenship, place of birth, childhood residence or migration history. Domestic education can occur for foreign-born individuals, and international education can occur for South African-born individuals.

Administrative missingness

Country was unresolved for 807 profiles (18.62%). Missingness may differ by role, department or profile-maintenance practice, so specified-profile percentages need not represent the entire staff population.

Heuristic country mapping

The mapping is keyword-based and was not accompanied by a manually labeled validation sample. Ambiguous institutional names, abbreviations, historical names and substring matches can generate false positives or false negatives.

Backward propagation assumption

Assigning a later parenthetical institution to earlier unparenthesized degrees may be correct for compact CV conventions, but it can misassign records when only the later credential belongs to that institution.

Full-string fallback

When the selected institution did not resolve, the parser searched the entire qualification string. A country associated with a later credential can therefore be assigned to the selected earlier credential.

Unnormalized organizational labels

The snapshot contained 204 designation strings and 313 department strings. Exact-label analysis avoids subjective recoding but splits spelling and punctuation variants.

Snapshot and extraction limits

Public profiles change over time, and the report did not rerun or independently audit the live web scrape. Empty biography and ORCID fields indicate that parts of the enrichment procedure did not yield usable data in the supplied file.

Recommended validation before external publication

Draw a stratified manual sample of domestic, international and unspecified records; verify the selected credential, institution and country against the source profile; report precision and coverage; and revise ambiguous keyword rules before making institutional or policy claims.

4. Conclusion

Within the 3,526 profiles that received a country classification, 87.55% were classified as South African first education and 12.45% as international. The international group was concentrated in UK and US country categories and varied substantially across selected designations and departments. Because the classifier is heuristic and 18.62% of profiles were unresolved, the results are best used as an auditable descriptive baseline and a guide for subsequent manual validation, not as a direct estimate of staff nationality or migration status.

Appendix A. Reproducibility notes

Table A1. Supplied analysis package

File	Role
wits_staff_directory.csv	Raw directory export; 4,333 profile records.
wits_staff_education_enriched.csv	Final analysis file; 4,333 rows and 14 columns.
scrape_wits_staff.py	Paginated directory harvester.
enrich_wits_staff_directory.py	Profile qualification, ORCID and biography extractor.
qual_analyzer.py	South African and foreign keyword taxonomies.
first_education_analyzer.py	Credential parsing, propagation and country assignment.
apply_first_education.py	Batch application of first-education parsing.

Computation formulas

Table A2. Report calculations

Metric	Formula
Specified profiles	Domestic + International
International share, specified	International / Specified x 100
Department inclusion	Specified department records $\geq$ 30
School-term prevalence	Profiles matching the pre-tertiary keyword expression / 4,333 x 100

Data source

University of the Witwatersrand public people-search directory: <https://www.wits.ac.za/people/search/>. The report computations use the locally supplied snapshot rather than a fresh extraction.

Appendix B. Complete international country distribution

Table B1. All 43 international country categories

Country	N	% intl.	Country	N	% intl.
UK	166	37.81%	Botswana	3	0.68%
USA	85	19.36%	Ghana	3	0.68%
Nigeria	18	4.10%	Egypt	3	0.68%
DR Congo	12	2.73%	Spain	2	0.46%
Canada	12	2.73%	Zambia	2	0.46%
Germany	11	2.51%	Norway	2	0.46%
France	11	2.51%	Ireland	2	0.46%
Australia	10	2.28%	Finland	2	0.46%
Zimbabwe	10	2.28%	Austria	2	0.46%
New Zealand	9	2.05%	Singapore	1	0.23%
Netherlands	9	2.05%	Israel	1	0.23%
Belgium	7	1.59%	Argentina	1	0.23%
Italy	7	1.59%	Poland	1	0.23%
Mauritius	5	1.14%	Sudan	1	0.23%
China	5	1.14%	Japan	1	0.23%
Brazil	5	1.14%	Uganda	1	0.23%
Sweden	5	1.14%	Malaysia	1	0.23%
India	4	0.91%	Greece	1	0.23%
Pakistan	4	0.91%	Malawi	1	0.23%
Switzerland	4	0.91%	Cameroon	1	0.23%
Kenya	4	0.91%	Lesotho	1	0.23%
Russia	3	0.68%			

Source: Recomputed from *First_Education_Country* in *wits_staff_education_enriched.csv*.